

Standing Localization and the Origin of Physical Locality

A Bounded-Locus Theorem for Standing-Bearing Persistence

Amos Jay Maley

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Abstract

A preceding analysis established that realized persistence cannot be reconstruction-null inside a fixed admissible continuation regime. This paper derives the next physical role: the structural origin of locality. It asks whether standing-bearing persistence can remain global, diffuse, trace-level, hidden, or unbounded once standing-relevant perturbation and lawful restoration are admitted. It cannot. Retained standing under disturbance requires a bounded role locus over which degradation, correction, retained identity, failure, and lawful continuation are jointly evaluated.

The theorem claim is the Standing Localization Theorem: within a fixed admissible realized persistence regime, restorative continuation constraints require a bounded standing-localization equivalence class for standing-bearing consequence retention. The physical-role claim is that locality first enters as bounded standing-localization, before metric distance, topology, spacetime regions, field support, or particle occupancy are available.

The bounded locus is not assumed to be a point, spatial region, manifold neighborhood, physical body, field support, or spacetime event. It is the minimal continuation-sensitive equivalence class through which standing-bearing consequence is preserved under perturbation and admissible same-domain restoration. Later geometric, field-theoretic, and particle-locality structures are downstream specializations of this locality role, not primitives imported into the proof.

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1 Introduction

1.1 Theorem claim and physical role

The present theorem states the origin of locality at the role level. The exact structural result is bounded standing-localization:

$$\text{RealPers}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv_{\text{Bound}}} \neq \emptyset.$$

The physical-role result is stronger than a negative scope statement: locality is first forced as the bounded locus over which perturbation burden, restoration, identity-memory, retained standing, and failure can be jointly evaluated.

Thus the paper does not say that locality is absent until a metric or manifold is later supplied. It says that metric, topological, field-theoretic, and particle-local notions of locality are later formal specializations of the bounded standing-localization role derived here.

The preceding paper isolated the first structural floor for the present reconstruction sequence. In a fixed admissible continuation regime, a target already typed as carrying realized persistence cannot remain null with respect to its own persistence-sensitive reconstruction structure [1]. The resulting structure is minimal: non-vacuous continuation-sensitive reconstruction is required, but no topology, metricity, locality, field content, particle ontology, dynamics, or cosmological origin follows at that stage.

The present paper asks the next question. Once realized persistence is non-vacuous, can its standing remain everywhere-at-once, diffuse over an unbounded aggregate, merely trace-level, or unlocalized while still retaining standing-bearing consequence under perturbation and lawful correction? The answer is negative. Realized persistence that carries standing-bearing consequence through disturbance must evaluate degradation, correction, retained identity, and failure somewhere in the role-theoretic sense: over a determinate bounded locus internal to the fixed regime. The first locality-relevant structure is therefore not geometry or metric distance, but bounded standing-localization.

The paper does not derive spacetime as a manifold, Lorentzian metric, causal-cone structure, finite propagation law, field algebra net, or particle support. It proves a weaker claim: any later locality reconstruction that carries standing-bearing persistence under perturbation and lawful restoration must preserve bounded continuation-sensitive localization.

1.2 The question after non-vacuity

The preceding paper proves the implication

$$\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D.$$

That result blocks reconstruction-null realized persistence. It does not yet identify the structure of the non-vacuous reconstruction interior. The interior might, at first glance, appear to admit several forms:

- (i) global standing, where consequence is retained without a bounded locus;
- (ii) diffuse standing, where consequence is spread across an unbounded or indeterminate aggregate;
- (iii) trace-only standing, where records succeed one another without a bounded restoration target;
- (iv) hidden localization, where an inaccessible reservoir allegedly grounds localization;
- (v) default localization, where a representative is selected without admitted witness structure;
- (vi) bounded localization, where standing-relevant consequence factors through a determinate locus.

The theorem proved below says that only the last option can carry standing-bearing consequence under perturbation and lawful restoration inside the same fixed domain.

1.3 Main result

The cumulative step can be written as

$$\text{Previous:} \quad \text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D,$$

followed by

$$\text{Present:} \quad \text{NonVacRec}_D \wedge \text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv \text{Bound}} \neq \emptyset.$$

The first implication blocks reconstruction-null realized persistence. The second, proved here, shows that non-vacuous standing-bearing persistence under perturbation and lawful restoration cannot remain unlocalized.

The central theorem is:

$$\text{RealPers}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv \text{Bound}} \neq \emptyset.$$

In the perturbation-and-restoration form:

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RestCap}_D \Rightarrow [B_D]_{\equiv \text{Bound}} \neq \emptyset.$$

The class $[B_D]_{\equiv \text{Bound}}$ is a bounded standing-localization equivalence class. Its boundedness is not physical enclosure. It means that there is a fixed admissibility-relevant role locus over which perturbation burden, correction, identity-memory, retained standing, and failure are jointly evaluated.

The informal controlling sentence is:

Locality first enters not as geometry, distance, or metric structure, but as the bounded locus required for standing-bearing persistence to retain consequence under perturbation and lawful restoration.

1.4 Claim levels and downstream specializations

The theorem level proves that standing-bearing consequence under perturbation and lawful restoration requires a bounded continuation-sensitive localization class. The physical-role level identifies that class as the primitive role later occupied by locality in physical theories.

This result does not yet specialize locality into spacetime regions, metric balls, manifold charts, field supports, local algebra nets, particle positions, causal diamonds, or empirical signal domains. Those require additional structure. But the locality role itself is derived here: standing cannot be retained, damaged, restored, or fail-closed without a bounded locus over which those verdicts are evaluated.

All necessity claims are conditional structural claims internal to a fixed admissible realized persistence regime. No observer information, entropy, probability, possible-world ensemble, anthropic selection, or hidden-variable locality is used as a primitive.

1.5 Derived here and deferred

The paper maintains the following separation throughout.

Derived here	Deferred to later papers
Bounded standing-localization	Metric geometry
Perturbation-burden locality	Finite propagation
Restorative locality	Causal-cone structure
Continuation-sensitive loci	Spacetime topology
Role-boundedness	Spatial distance or diameter
Boundary-trace anchoring	Field support or local algebra nets

This distinction is essential. The argument derives the minimal bounded role structure required for standing-bearing persistence; it does not yet derive space.

1.6 Dependency order

The proof imports only the following structural dependencies from the AASC corpus. They are used as fixed-domain role constraints rather than empirical postulates.

Dependency	Role in this paper
Non-vacuous realized persistence	The preceding result gives $\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D$, blocking reconstruction-null persistence [1].
Continuation-locus induction	Non-degenerate standing-bearing continuation induces continuation-locus equivalence rather than mere trace succession [2].
Boundary-trace fixation	A continuation locus is role-anchored by admissible boundary traces, not by arbitrary representative selection [3].
Persistence-regime realization	Irreversible standing-bearing continuation induces a realized persistence-regime equivalence class [4].
Standing-localization theorem	Existing restoration-localization machinery supplies the bounded restorative-locality pressure refined here as the pre-metric locality role [5].
AMetric boundary discipline	No selector, metric, default, probability, parameter, or canonical representative can be imported without admitted witness structure [6].
Anchor-tensor-skin discipline	Standing-bearing role load is anchor or tensor content; skin changes presentation without carrying gate authority [7].
Fixed-domain exhaustion	Same-domain alternatives must preserve the target, gate, quotient, transport, failure, and role structure or else count as scope change, bookkeeping, or inadmissible import [8].

1.7 Terminology translation

The table records the intended structural roles of the most important terms.

Framework term	Structural role
Admissibility	Lawful same-domain admission, rejection, continuation, redescription, and fail-closed status.
Standing	Persistence-bearing admissibility status retained through licensed continuation.
Realized persistence	Actual retention of standing-bearing consequence, not mere capacity or description.
Perturbation	A standing-relevant disturbance that can alter retained consequence unless lawful correction preserves it.
Restorative continuation constraints	Lawful same-domain restoration constraints distinguishing degradation, correction, retained identity, standing preservation, and failure.
Bounded identity locus	Minimal same-domain locus over which perturbation burden, restoration, identity-memory, retained standing, and failure are jointly evaluated.

Framework term	Structural role
Localization	Factorization of standing-bearing persistence through bounded continuation-sensitive equivalence classes.
Pre-metric locality	Bounded standing localization without primitive distance, topology, manifold structure, causal cones, or field support.

2 Fixed-domain preliminaries

This section fixes the target type. The definitions are intentionally weak. They do not assume metric, topological, causal, spatial, physical, or informational structure.

Definition 2.1 (Fixed admissible continuation regime). A fixed admissible continuation regime D is a comparison setting in which candidates are evaluated with respect to a stable admissibility gate, lawful same-domain continuation, standing-bearing status, quotient identity, role participation, redescription discipline, and fail-closed treatment of standing failure. Changing the gate, target vocabulary, failure semantics, comparison relation, or evaluated fragment changes the regime.

Definition 2.2 (Continuation history). A continuation history $h \in \mathbf{Hist}_D$ is a finite or schematically finite same-domain record of admissibility-relevant continuation acts, perturbations, restorations, failures, standing retention, and lawful redescriptions in D . A history is not a temporal sequence unless later structure supplies a temporal interpretation.

Definition 2.3 (Standing-bearing persistence). $\mathbf{RealPers}_D$ holds when a minimally realized regime actually carries standing-bearing consequence through lawful same-domain continuation. Standing denotes the admissibility-preserving persistence status required for continuation-sensitive burden assignment and restoration evaluation; it distinguishes retained consequence from failure, replacement, external description, or skin-level trace.

Definition 2.4 (Persistence interior from the preceding paper). The preceding paper defines reconstruction classes $\mathcal{R}_D = \mathbf{Hist}_D / \equiv_{\text{Rec}}$ induced by admissibility-preserving persistence-sensitive reconstruction maps and proves that realized persistence requires non-identical standing-relevant reconstruction classes. We write the resulting minimal non-vacuous interior as $[\mathbf{Int}_D]_{\equiv_{\text{Int}}} \neq \emptyset$. This interior is not yet a locality structure.

Remark 2.5 (Why non-vacuity is not locality). A non-vacuous reconstruction interior can distinguish retention from failure or lawful continuation from replacement without yet supplying bounded localization. The preceding paper blocks total reconstruction collapse; this paper adds perturbation and restoration pressure to derive bounded loci. The two steps must not be conflated.

Definition 2.6 (Standing-relevant perturbation). $\mathbf{Perturb}_D$ holds when the realized persistence regime admits a disturbance profile capable of altering standing-bearing continuation unless admissibility-preserving boundary, restoration, or localization structure preserves the relevant standing consequence. Perturbation is not empirical noise. It is standing-relevant disturbance.

Definition 2.7 (Restoration capacity). $\mathbf{RestCap}_D$ holds when standing-relevant consequence degradation admits lawful continuation-preserving correction without identity reset, illicit same-act correction, unrestricted global reindexing, hidden selector import, or target-domain drift.

Definition 2.8 (Restorative continuation constraints). RestCon_D is the constraint profile by which perturbation, degradation, restoration, retained identity, bounded restorative locality, standing preservation, and failure are distinguished inside the same fixed regime. These constraints are required whenever standing is preserved through lawful restoration rather than by inert classification, representative choice, hidden selector, identity reset, or global reindexing.

Definition 2.9 (Bounded identity locus). A bounded identity locus B_D is the minimal same-domain standing-relevant locus over which perturbation burden, correction, identity-memory, retained standing, and failure are jointly evaluated. Boundedness means boundedness relative to the admissibility-relevant restoration and evaluation domain. It does not mean spatial enclosure, compactness, finite cardinality, metric boundedness, physical extension, material embodiment, or coordinate diameter.

Remark 2.10 (Bounded relative to what). Throughout this paper, “bounded” means role-bounded: the relevant perturbation, correction, retained identity, and failure verdict are evaluated over a delimited continuation-sensitive locus rather than over an unbounded global whole, an indeterminate aggregate, or an external selector. No metric bound, topological compactness, or spatial size is assumed.

Definition 2.11 (Standing localization). Standing localization holds when standing-bearing persistence factors through bounded continuation-sensitive equivalence classes. A bounded standing-localization structure B_D carries the role profile required for retained consequence under perturbation: boundary, identity-memory, perturbation burden, restorative continuation constraints, quotient stability, and fail-closed status.

Definition 2.12 (Standing-localization equivalence on histories). For histories $h, k \in \text{Hist}_D$, write $h \equiv_{\text{sLoc}} k$ iff their standing-relevant perturbation burdens, restoration obligations, identity-memory roles, retained-standing verdicts, quotient behavior, and fail-closed outcomes remain jointly admissibly reconstructible as the same standing-bearing lineage without identity reset, external selector import, domain drift, or fail-open relabeling. A bounded localization class is an equivalence class of histories under this relation restricted to a common bounded identity locus.

Definition 2.13 (Bounded-localization equivalence). Let X and Y be candidate continuation loci, histories, or bounded standing-localization structures in D . We write $X \equiv_{\text{Bound}} Y$ iff their standing-relevant perturbation burdens and restoration burdens remain jointly admissibly reconstructible without identity reset, external selector import, fail-open relabeling, or domain drift. Equivalently, they preserve the same boundary role, identity-memory profile, perturbation burden, retained restorative continuation constraints, standing preservation status, transport behavior, quotient behavior, and fail-closed failure behavior under lawful same-domain redescription.

Remark 2.14 (No primitive topology). The words “locus”, “boundary”, and “locality” are not primitive topological vocabulary here. They name admissibility-bearing role constraints. No open sets, neighborhoods, charts, metric balls, causal diamonds, spatial regions, or algebraic nets are assumed.

3 From non-vacuity to localization pressure

The first paper gives the non-vacuous interior needed for realized persistence. The present paper asks which further structure is forced when that interior is exposed to perturbation and restoration.

Proposition 3.1 (Non-vacuity floor). *For any fixed admissible continuation regime D ,*

$$\text{MinReal}_D \wedge \text{RealPers}_D \Rightarrow \text{NonVacRec}_D.$$

Consequently realized persistence has at least one non-trivial continuation-sensitive reconstruction distinction.

Proof. This is the Reconstruction Non-Vacuity result of the preceding paper [1]. It is imported here only as a floor: it supplies non-vacuous reconstruction, not localization. $\square$

Proposition 3.2 (Localization pressure is not supplied by non-vacuity alone). *NonVacRec_D does not by itself imply $[B_D]_{\equiv_{\text{Bound}}} \neq \emptyset$.*

Proof. A reconstruction distinction may distinguish standing retention from standing failure, lawful continuation from replacement, or role participation from skin without specifying the bounded locus over which perturbation burden and correction are jointly evaluated. Such a distinction is non-vacuous but not yet localized. Therefore localization requires additional role pressure. $\square$

Definition 3.3 (Standing consequence retention). Standing consequence retention holds in D when a continuation lineage preserves an admissibility-bearing consequence across licensed continuation rather than merely leaving a trace, syntax, semantic label, or external record.

Lemma 3.4 (Consequence retention requires evaluable difference). *If standing consequence is retained, the regime must distinguish retained consequence from loss, replacement, trace succession, and skin-level redescription.*

Proof. If retained consequence and loss are not distinguished, then the status of the lineage is reconstruction-null with respect to standing. If replacement and continuation are not distinguished, then identity retention is unavailable. If trace succession and standing consequence are not distinguished, then a record can substitute for persistence. Each case contradicts realized persistence as used here and as established in the preceding paper. $\square$

The key new pressure is perturbation. A regime can be non-vacuous without yet being asked where a disturbance is charged or where correction is evaluated. Once perturbation is standing-relevant, however, retention becomes a locus problem.

4 Global standing failure

A global standing claim says that standing-bearing consequence persists under perturbation without any bounded standing-positive localization structure supporting the relevant consequence. The term “global” here does not mean large spatial scale. It means unbounded role-distribution without a determinate identity locus.

Definition 4.1 (Global standing). GlobalStand_D holds when standing is claimed to persist under perturbation without a bounded identity locus through which perturbation burden, restoration, identity-memory, retained consequence, and failure are jointly evaluated.

Lemma 4.2 (Global standing indeterminacy).

$$\text{GlobalStand}_D \wedge \text{Perturb}_D \Rightarrow \neg \text{RestLoc}_D.$$

A global standing claim under perturbation supplies no determinate target for restorative-local evaluation.

Proof. A standing-relevant perturbation changes or threatens standing-bearing consequence. To evaluate whether standing is retained, the regime must determine what is disturbed, what is corrected, which lineage is preserved, and when failure occurs. A global standing claim denies the bounded locus on which those assignments are made. The alleged restoration then becomes one of four things: unrestricted global reindexing, envelope import, external transfer, or inert bookkeeping. Reindexing and envelope import alter the target; external transfer violates same-domain standing conservation unless freshly gated; bookkeeping changes no standing verdict. Hence global standing under perturbation cannot supply restorative locality. $\square$

Corollary 4.3 (Global standing exclusion).

$$\text{GlobalStand}_D \wedge \text{Perturb}_D \wedge \text{RestCap}_D \Rightarrow \neg \text{RealPers}_D$$

for the original same-domain target.

Proof. Restoration capacity requires lawful correction that preserves standing-bearing identity rather than resetting or globally reindexing it. The previous lemma shows that global standing under perturbation lacks restorative locality. With restoration capacity claimed, the mismatch can only be resolved by scope change, hidden selection, identity reset, or denial of the target phenomenon. None preserves the original same-domain realized persistence target. $\square$

Remark 4.4 (Not a denial of global descriptions). A global description may be a useful representation inside a later admitted regime. The point is narrower: global description does not by itself carry standing-bearing restorative locality. If a global formalism supplies local roles through admitted structure, then the locality is in that admitted structure, not in an unlocalized global standing claim.

5 Diffuse and trace-only standing failure

Diffuse standing is different from global standing. A global claim denies bounded locus structure by assigning standing to the whole regime. A diffuse claim distributes standing across an unbounded or indeterminate aggregate without a determinate locus charged with perturbation and restoration.

Definition 5.1 (Diffuse standing). DiffuseStand_D holds when standing-bearing consequence is distributed over an unbounded, indeterminate, or non-local aggregate without a bounded identity locus carrying perturbation burden, restorative continuation constraints, identity-memory, standing margin, and fail-closed status.

Lemma 5.2 (Diffuse standing collapse).

$$\text{DiffuseStand}_D \wedge \text{Perturb}_D \Rightarrow \neg \text{StableConseqRet}_D.$$

Diffuse standing under perturbation cannot stably retain the same standing-bearing consequence.

Proof. If standing is diffuse, no bounded locus is responsible for the disturbance. Any attempted correction must either restore nothing determinate, restore the entire aggregate globally, import a hidden selector to choose the affected locus, reset identity after the disturbance, or change the domain to a larger envelope. Restoring nothing determinate is not restoration. Global restoration is unrestricted reindexing. Hidden selection violates AMetric no-default discipline unless admitted as non-null governance. Identity reset loses the original standing lineage. Envelope drift changes the target. Therefore diffuse standing cannot retain the same standing-bearing consequence under perturbation. $\square$

Definition 5.3 (Trace-only standing). TraceStand_D holds when a sequence of traces, records, labels, or descriptions is offered in place of standing-bearing consequence retention through a bounded locus.

Lemma 5.4 (Trace-only insufficiency).

$$\text{TraceStand}_D \wedge \text{Perturb}_D \wedge \text{RestCap}_D \Rightarrow \neg \text{RealPers}_D$$

for the target lineage.

Proof. A trace can record that a status was assigned, that a perturbation occurred, or that a description changed. But restoration capacity requires correction of standing-relevant degradation as the same lineage. A trace-only regime supplies no bounded target of disturbance, no identity-memory criterion, no restoration throughput, and no fail-closed distinction between restored retention and post-hoc relabeling. Thus trace-only succession cannot replace standing-bearing restoration. $\square$

Corollary 5.5 (Diffuse and trace-only exclusion). *If RealPers_D , Perturb_D , and RestCap_D are all in force, standing cannot be diffuse or trace-only.*

Proof. Diffuse standing fails stable consequence retention. Trace-only standing fails same-lineage restoration. Both conflict with realized persistence under perturbation and restoration capacity. $\square$

6 Perturbation burden and restorative continuation constraints

The core constructive step is to show that perturbation and restorative continuation constraints force bounded loci. This is the transition from non-vacuous persistence to pre-metric locality.

Definition 6.1 (Perturbation burden). A perturbation burden is the continuation-sensitive consequence assignment induced when a disturbance threatens or degrades the retained standing of a continuation lineage. It is not a force, energy, noise level, entropy increase, or empirical error unless later physical structure supplies those interpretations.

Proposition 6.2 (Perturbation burden requires locus). *If a perturbation is standing-relevant, then there must be a determinate locus to which its burden is assigned, unless the perturbation is inert for standing.*

$$\text{Perturb}_D \wedge \text{standing-relevant}(p) \Rightarrow \exists B_D \text{ burdened}(p, B_D).$$

Proof. A perturbation is standing-relevant only if it can alter retained standing, failure status, restoration requirement, identity-memory, role participation, or quotient behavior. If no locus bears the perturbation burden, then no same-domain criterion determines what has degraded, what requires correction, or whether correction preserves the same lineage. The perturbation either changes no standing verdict, in which case it is inert, or it changes standing without an evaluable target, in which case persistence-sensitive reconstruction fails. Therefore a standing-relevant perturbation requires a determinate locus. $\square$

Proposition 6.3 (Restoration comparison requires lineage). *Restorative continuation constraints require comparison of at least three role states: pre-perturbation standing, perturbed standing, and restored standing, all typed as belonging to the same candidate lineage unless failure is declared.*

Proof. Without the pre-perturbation state there is no retained consequence to preserve. Without the perturbed state there is no standing-relevant degradation. Without the restored state there is no restoration. Without a same-lineage criterion, the restored state may be replacement, envelope import, or external transfer. Restorative continuation constraints are precisely the role structure distinguishing these cases. $\square$

Proposition 6.4 (Restorative Continuation Locality). *If perturbation and restoration are not evaluated over bounded continuation-sensitive loci, then standing-burden assignment, persistence retention, and identity-preserving correction become indeterminate. Consequently restorative continuation constraints force localization in the pre-metric role sense.*

Proof. Standing-burden assignment requires a determinate target of degradation. Persistence retention requires a determinate lineage whose standing is retained rather than replaced. Identity-preserving correction requires a comparison between pre-perturbation, perturbed, and restored states as the same lineage. Without a bounded continuation-sensitive locus, each of these requirements fails or is supplied by a global reindexing, hidden selector, identity reset, or domain change. Therefore restorative continuation constraints cannot be evaluated without localization. $\square$

Theorem 6.5 (Bounded locus requirement).

$$\text{RestCon}_D \Rightarrow \exists B_D.$$

Restorative continuation constraints require a bounded identity locus over which perturbation burden, correction, identity-memory, retained standing, and failure are distinguished.

Proof. Restorative continuation constraints compare disturbed and restored continuation states as the same standing-bearing lineage unless fail-closed status is reached. That comparison requires a boundary fixing what is being restored, an identity-memory criterion fixing whether the restored lineage is the same lineage, a perturbation-burden assignment fixing where the disturbance burden is assigned, a correction role fixing the lawful restoration contribution, and a failure condition fixing when standing is not retained. These roles cannot be evaluated over an unbounded global whole, a diffuse aggregate, a trace-only record, an external selector, or a post-hoc identity reset without losing same-domain restorative continuation constraints. Therefore restorative continuation constraints require a bounded identity locus. $\square$

Corollary 6.6 (Restorative locality).

$$\text{RestCon}_D \Rightarrow \text{RestLoc}_D.$$

Proof. By the bounded locus requirement, restorative continuation constraints factor through a bounded identity locus. Restoration is local in the pre-metric role sense: it is evaluated on the locus whose standing-bearing identity is being preserved or lost. $\square$

7 Standing Localization Theorem

We can now state the main theorem in both the restorative-constraint and perturbation forms.

Theorem 7.1 (Standing Localization). *Let D be a fixed admissible continuation regime. If D supports realized persistence with restorative continuation constraints, then standing-bearing consequence cannot remain global, diffuse, trace-only, or unlocalized. It induces a bounded standing-localization equivalence class:*

$$\text{RealPers}_D \wedge \text{RestCon}_D \Rightarrow [B_D]_{\equiv_{\text{Bound}}} \neq \emptyset.$$

Proof. Assume $\text{RealPers}_D \wedge \text{RestCon}_D$. Realized persistence supplies standing-bearing consequence retention inside a fixed domain. Restorative continuation constraints supply the role distinction between degradation, correction, retained identity, standing preservation, and fail-closed failure. By the bounded locus requirement, those distinctions require a bounded identity locus B_D . Because the locus carries perturbation burden, identity-memory, restorative continuation constraints, standing preservation, quotient behavior, and fail-closed status, it is not skin-only presentation. Because it is fixed by standing-relevant roles rather than representative selection, it is not an AMetric default. Because it preserves the same role profile under lawful redescription, it determines an equivalence class under $\equiv_{\text{Bound}}$. Therefore $[B_D]_{\equiv_{\text{Bound}}} \neq \emptyset$. $\square$

Corollary 7.2 (Perturbation-and-restoration form).

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RestCap}_D \Rightarrow [B_D]_{\equiv_{\text{Bound}}} \neq \emptyset.$$

Proof. Realized persistence supplies consequence retention. A standing-relevant perturbation threatens that retention. Restoration capacity requires lawful correction without identity reset, same-act correction, global reindexing, hidden selection, or domain drift. Together these induce restorative continuation constraints. The Standing Localization Theorem then yields the bounded localization class. $\square$

Corollary 7.3 (Pre-metric locality inheritance). *Any later spacetime-local reconstruction of D must inherit a pre-metric bounded standing-localization role if it is to carry realized standing-bearing persistence under perturbation and lawful restoration.*

Proof. Later spacetime structure may add metric, topological, causal, field-theoretic, or dynamical roles. But if it carries realized persistence under perturbation and restoration, the Standing Localization Theorem already requires bounded standing localization. Therefore spacetime locality, if reconstructed later, must be compatible with this pre-metric locality role. $\square$

8 Equivalence, uniqueness, and no representative selection

The theorem induces an equivalence class, not a canonical representative. This distinction is essential. A representative selector would add governance not licensed by the localization theorem.

Definition 8.1 (Localization profile). The localization profile of a bounded identity locus B_D is the tuple

$$\Pi(B_D) = \left(\begin{array}{l} \text{boundary, identity-memory, perturbation burden,} \\ \text{restorative continuation constraints, standing status,} \\ \text{quotient behavior, failure behavior} \end{array} \right).$$

Two loci are bounded-localization equivalent when their profiles agree up to lawful same-domain redescription.

Proposition 8.2 (Uniqueness up to bounded-localization equivalence). *If B_D and B'_D preserve the same localization profile under lawful same-domain redescription, then*

$$B_D \equiv_{\text{Bound}} B'_D.$$

Proof. The localization profile lists the admissibility-bearing roles of standing localization. If two loci preserve all of them, no remaining same-scope standing-relevant difference is available. Any further difference is presentation, bookkeeping, representative choice, or scope change. Therefore they are equivalent under $\equiv_{\text{Bound}}$. $\square$

Corollary 8.3 (No canonical representative).

$$[B_D]_{\equiv_{\text{Bound}}} \neq \emptyset \not\equiv \exists! B_D.$$

A bounded standing-localization equivalence class need not select a unique representative.

Proof. Representative selection requires a selector, index, metric, parameter, coordinate rule, or external convention with standing-bearing authority. None is supplied by bounded-localization equivalence itself. If such structure is added, it is additional admitted governance or a later reconstruction layer. $\square$

9 Boundary trace and no default localization

Bounded localization must be fixed by admitted continuation and boundary-trace structure. It cannot be selected by a default rule at the AMetric boundary.

Definition 9.1 (Boundary-trace fixation). Boundary-trace fixation holds when a continuation locus is role-anchored by admissible boundary traces: stable standing-relevant interface conditions that fix what counts as the same locus under lawful continuation. Boundary trace fixes a role class. It does not select a unique representative unless further selector structure is admitted.

Proposition 9.2 (Boundary-trace role anchoring). *If a continuation locus is non-degenerate and admitted boundary traces fix its role participation, then the induced localization class is role-anchored rather than freely selected:*

$$[L_D]_{\equiv_{\text{Loc}}} \wedge \text{BT}_D \Rightarrow [B_D]_{\equiv_{\text{Bound}}} \text{ is role-anchored.}$$

Proof. Continuation-locus induction supplies a locus-equivalence class rather than a bare trace sequence. Boundary-trace fixation supplies role anchors for continuation across admissible interfaces. When restorative continuation constraints require a bounded identity locus, that locus must preserve the boundary role over which perturbation and correction are evaluated. Thus the localization class is fixed by role participation, not by representative choice. $\square$

Theorem 9.3 (No Default Locus).

$$\text{AMetric}_D \Rightarrow \neg \text{DefaultLoc}_D(B_D)$$

unless additional admissibility-bearing selector structure is introduced. If such structure is introduced, the localization is no longer default.

Proof. A default localization rule selects a locus, representative, scale, region, parameter, or coordinate as standing-bearing without deriving it from admitted standing-relevant structure. The AMetric boundary supplies no selector, metric, ranking, default, probability, canonical representative, or transport rule. If witness structure is supplied, that witness is non-null admitted governance; if it is not supplied, no admissibility-bearing localization occurs. Therefore default localization is unavailable. $\square$

Corollary 9.4 (No arbitrary spacetime point import). *The theorem does not license the importation of primitive spacetime points, metric regions, coordinate charts, or canonical representatives as default localizers. Such structures must be introduced by later admitted reconstruction work or not at all.*

10 Hidden reservoirs and external selectors

A common escape route says that standing localization can be grounded by a hidden latent structure while no bounded locus appears in the admitted continuation regime. This route fails for the same reason hidden reconstruction reservoirs fail in the preceding paper: if the hidden structure is inaccessible to admissible continuation-sensitive roles, it is inert; if it is accessible, the access relation is non-null governance.

Definition 10.1 (Hidden localization reservoir). HiddenLoc_D holds when a latent structure is alleged to ground standing localization while remaining inaccessible to all admissibility-preserving continuation, restoration, boundary-trace, quotient, and reconstruction roles inside D .

Theorem 10.2 (Hidden Localization Reservoir Exclusion). *A hidden localization reservoir cannot ground realized standing localization for D . If it remains inaccessible to admissible continuation-sensitive roles, it is realization-inert for D . If it becomes accessible, the access relation is non-null governance or a scope change.*

Proof. If the alleged reservoir is inaccessible to the roles that determine perturbation burden, restoration, identity-memory, boundary trace, standing retention, failure, quotient behavior, and lawful redescription, then it makes no standing-relevant difference inside D . It cannot ground realized localization. If it does make such a difference, there must be an access relation, witness, selector, memory, role anchor, transport rule, quotient rule, or boundary trace connecting it to D . That connector is admitted non-null governance unless the domain changes. Hence a hidden inaccessible reservoir does not rescue unlocalized standing. $\square$

Definition 10.3 (External selector localization). ExtSel_D holds when a locus is chosen by an external semantic rule, observer convention, coordinate convention, ensemble index, or representative selector not licensed by the fixed domain.

Lemma 10.4 (External selectors are not same-domain localization).

$$\text{ExtSel}_D \Rightarrow \text{bookkeeping, imported governance, or domain change.}$$

Proof. If an external selector changes no standing verdict, it is bookkeeping. If it changes a standing verdict while preserving the same target, it is imported governance and must be admitted as non-null structure. If it changes the target vocabulary, gate, comparison relation, or evaluated fragment, it is domain change. None gives default same-domain standing localization. $\square$

11 Skin, anchor, and tensor load

Standing localization carries role load. It cannot be only a presentation.

Proposition 11.1 (Localized standing is not skin).

$$[B_D]_{\equiv_{\text{Bound}}} \neq \emptyset \Rightarrow \text{Anchor}(B_D) \vee \text{Tensor}(B_D).$$

Consequently

$$\text{SkinOnly}(B_D) \Rightarrow \neg \text{LocalizedStanding}_D.$$

Proof. A bounded standing-localization structure preserves perturbation burden, identity-memory, restorative continuation constraints, boundary role, quotient behavior, standing retention, and fail-closed failure. These roles affect admissibility-bearing continuation. By anchor–tensor–skin discipline, admissibility-bearing authority is anchor or tensor load; skin only changes presentation. Therefore localized standing is not skin-only. $\square$

Remark 11.2 (Why this matters for spacetime language). A diagram, coordinate chart, metric notation, or region label can display locality without carrying it. The theorem concerns the role structure that must be preserved under perturbation and restoration. Later spacetime representations may be excellent skins over that structure, but the localization role is not supplied by the skin.

12 Finite witness model

The finite model is not a physical model. It instantiates the theorem architecture in a finite setting.

Let $G = (V, E)$ be a finite continuation graph with stages

$$V = \{s_0, s_1, s'_1, f\}.$$

The intended standing lineage is

$$s_0 \xrightarrow{p} s_1 \xrightarrow{\rho} s'_1,$$

where p is a standing-relevant perturbation and ρ is a lawful restoration map when it is admissible. The vertex f represents fail-closed standing loss. The basic restoration square is

$$\begin{array}{ccccc} s_0 & \xrightarrow{p} & s_1 & \xrightarrow{\rho} & s'_1 \\ & & \downarrow & & \\ & & f & & \end{array}$$

and quotient discipline is expressed schematically by

$$\begin{array}{ccc} \text{Hist}_G & \xrightarrow{\rho} & \text{Hist}_G \\ \pi_{\equiv \text{Bound}} \downarrow & & \downarrow \pi_{\equiv \text{Bound}} \\ \text{Hist}_G / \equiv_{\text{Bound}} & \xrightarrow{\bar{\rho}} & \text{Hist}_G / \equiv_{\text{Bound}} . \end{array}$$

The upper row records the perturbation-restoration lineage. The lower row records the requirement that restoration remain well-defined after bounded-localization quotienting; if the quotient collapses restored standing and fail-closed loss, the restoration claim is not well-defined.

Let

$$J : V \rightarrow \{0, 1\}, \quad R : V \rightarrow \{0, 1\}, \quad D_p : V \rightarrow \{0, 1\}$$

record identity-memory, restorative throughput, and perturbation burden. Let $\sim$ be quotient equivalence under lawful redescription, and let $\text{Stand}(v) \in \{0, 1\}$ be standing status.

A localization candidate is an equivalence class $B \subseteq V / \sim$ with profile

$$\Pi(B) = (J_B, R_B, D_B, \text{Stand}_B, \text{Fail}_B).$$

For classes $B, C \subseteq V / \sim$, define $B \equiv_{\text{Bound}} C$ exactly when their identity-memory, perturbation-burden, restoration-throughput, standing-status, and fail-closed verdicts are jointly reconstructible under the same lawful restoration lineage without identity reset, external selection, or quotient collapse. Thus the finite model instantiates the abstract bounded-localization equivalence relation rather than merely illustrating it. The standing margin notation may be written

$$\Sigma_B = J_B R_B - D_B,$$

not as an empirical scalar but as a compressed verdict form: retained identity times local restoration, evaluated against perturbation burden.

Theorem 12.1 (Finite localization witness). *In the model G , if realized persistence, standing-relevant perturbation, and lawful restoration capacity are all present, then there exists a bounded localization class:*

$$\text{RealPers}(G) \wedge \text{Perturb}(G) \wedge \text{RestCap}(G) \Rightarrow [B_G]_{\equiv_{\text{Bound}}} \neq \emptyset.$$

Proof. Consider the alternatives. If all vertices collapse into one global standing class, then s'_1 and f cannot be distinguished with respect to restored retention versus fail-closed loss; restoration is indistinguishable from failure. If the burden is diffuse across an indeterminate subset of V , then no class is responsible for the transition from s_0 to s_1 or for the restoration transition from s_1 to s'_1 ; the burden is either inert, globalized, or selector-dependent. If every vertex is independent without a same-lineage locus, then s_0, s_1, s'_1 cannot be reconstructed as the same standing lineage; restoration is replacement. If perturbation burden is not assigned to any class, then p is either inert or destroys persistence-sensitive reconstruction. If restoration is supplied by an external selector, the selector is imported governance or domain change. If restoration resets identity, s'_1 is not restored retention of s_1 . If quotient discipline is dropped, presentation differences can manufacture fake standing distinctions. If fail-closed status is dropped, failed standing can be relabeled as restored standing.

The only surviving structure is a quotient class that jointly carries identity-memory, perturbation burden, local restoration, standing status, quotient stability, and fail-closed behavior. That is precisely a bounded standing-localization class. $\square$

Corollary 12.2 (Witness synthesis). *The finite witness instantiates the general theorem architecture of Proposition 6.4, Theorem 6.5, and Theorem 7.1: global collapse destroys restoration discrimination; diffuse burden destroys responsibility assignment; total independence destroys lineage; hidden selection violates same-domain discipline; identity reset destroys retained standing; and bounded localization is the remaining same-domain role structure.*

13 Exhaustion and normal form

The preceding results can be stated as a compact exclusion. Let an unlocalized persistence route be a same-domain strategy that attempts to preserve RealPers_D under Perturb_D and RestCap_D while denying $[B_D]_{\equiv_{\text{Bound}}} \neq \emptyset$.

Theorem 13.1 (Localization route exhaustion). *Let X be any candidate same-domain regime offered as realized standing-bearing persistence under perturbation and restorative capacity. After normalization by target, access, and role import, exactly one of the following classes applies:*

E1. Bounded localization: X preserves a bounded standing-localization class.

- E2.** Restoration-indeterminate continuation: *standing is global, diffuse, trace-only, or otherwise lacks a determinate restoration target.*
- E3.** Hidden or default localization: *localization is supplied by an inaccessible reservoir, AMetric default, arbitrary representative, or external selector.*
- E4.** Identity-reset continuation: *restoration is replaced by identity reset, same-act correction, fail-open relabeling, or replacement of the lineage.*
- E5.** Non-realized continuation: *the target, gate, quotient, boundary trace, or domain changes, so the alleged route no longer preserves the same realized persistence claim.*

Only **E1** remains a same-domain realized-persistence candidate under perturbation and restorative capacity. Therefore

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RestCap}_D \Rightarrow [B_D]_{\equiv_{\text{Bound}}} \neq \emptyset.$$

Proof. Either the candidate preserves the roles of bounded localization or it does not. If it does, **E1** holds and the conclusion follows. If it does not, the failure is restoration-indeterminate, hidden/default-selected, identity-resetting, or non-realized. Global, diffuse, and trace-only standing are restoration-indeterminate because perturbation burden and retained standing cannot be assigned to a determinate lineage. Hidden reservoirs, AMetric defaults, arbitrary representatives, and external selectors do not supply same-domain localization unless their governance is admitted; if admitted, they are no longer hidden or default. Identity resets and fail-open relabelings replace restoration with lineage loss. Domain drift changes the target rather than preserving it. No additional category exists outside this partition: any alleged further candidate either preserves bounded continuation-sensitive restoration roles and is therefore **E1**, changes no standing verdict and is inert, imports role load and must be admitted as governance, or changes the target. Hence only bounded standing localization survives. $\square$

Theorem 13.2 (Bounded-localization normal form). *Every admissibility-preserving localization regime carrying realized persistence under perturbation and restorative capacity factorizes through:*

- (a) *a non-vacuous reconstruction interior;*
- (b) *standing-bearing consequence retention;*
- (c) *perturbation-burden assignment;*
- (d) *restorative continuation constraints;*
- (e) *a bounded identity locus with boundary, identity-memory, restoration, perturbation burden, quotient stability, and fail-closed failure;*
- (f) *boundary-trace fixation and no default representative selection;*
- (g) *bounded-localization equivalence up to lawful same-domain redescription.*

Thus

$$\text{RealPers}_D \wedge \text{Perturb}_D \wedge \text{RestCap}_D \Rightarrow ([P_D]_{\equiv_{\text{Pers}}}, \text{RestCon}_D, \text{RestLoc}_D, [B_D]_{\equiv_{\text{Bound}}}).$$

Proof. Realized persistence supplies retained standing consequence. Perturbation exposure tests whether consequence can survive disturbance. Restoration capacity supplies restorative continuation constraints. These constraints require bounded restorative locality. Boundary-trace and quotient discipline fix what the locus preserves under redescription. A Metric non-selection blocks default representatives. Skin exclusion blocks purely presentational localization. The resulting structure is exactly the listed normal form. $\square$

14 Relation to later locality reconstructions

The result concerns a pre-metric localization role, not a completed derivation of spacetime. It identifies the bounded role that any later locality reconstruction must play if it is to carry realized persistence. It does not identify that role with any mature spacetime structure.

14.1 Pre-metric locality

Pre-metric locality is bounded standing localization without primitive distance. It is weaker than any of the following:

- topological locality by open neighborhoods;
- metric locality by distance or bounded balls;
- relativistic locality by causal cones or spacelike separation;
- field-theoretic locality by supports or local algebras;
- particle locality by object occupancy;
- operational locality by measurement settings and outcomes.

The theorem says only that standing-bearing persistence under perturbation and lawful restoration requires bounded loci. A later metric theory may represent these loci as regions, neighborhoods, events, or algebraic domains. Such representation is additional structure.

14.2 Why this is a pre-metric locality theorem

One might object that without distance the result is not locality. That objection presupposes a later, stronger notion. The present result concerns the minimal role common to those stronger notions: a standing-relevant distinction between what belongs to the evaluated locus and what does not. Without such role-boundedness, no perturbation burden can be assigned, no restoration can be evaluated, no identity can be retained, and no failure can be declared as the failure of that lineage. In that limited sense, pre-metric localization precedes metric locality.

14.3 Compatibility with nonlocal tensor structure

The theorem concerns realized standing localization, not total tensor-factor separability and not Bell-local hidden-variable factorization. A regime may contain global compatibility constraints, boundary-level tensor structure, or nonlocal correlation structure while still requiring bounded loci for standing-bearing perturbation and restoration. Conversely, deriving bounded standing-localization does not imply that the total theory factors into independent local components. The locality proved here is restoration/evaluation locality for standing-bearing persistence.

14.4 Relation to algebraic and operational locality

Algebraic quantum field theory begins with assignments of algebras to spacetime regions and studies isotony, covariance, locality, and related structural constraints [10, 11]. Operational and generalized-probabilistic reconstructions often begin with information-processing, measurement, or preparation-transformation primitives [12, 13, 14, 15]. The present analysis begins at a different structural level: it does not assume regions, algebras, observers, measurements, information, preparations, or probabilities. It asks what bounded role structure is required for standing-bearing persistence before such frameworks are introduced. This does not compete with those frameworks; it specifies a prior admissibility role that any persistence-bearing local framework must satisfy.

14.5 Quantum locality and Bell scope

The result should not be read as a local-hidden-variable theorem. Bell-type constraints show that quantum correlations cannot be classified as local hidden-variable factorizations under the usual assumptions [16, 17, 18]. The present theorem does not alter that result. It distinguishes bounded standing localization from hidden-variable locality.

The theorem is also compatible with globally entangled, tensorially non-factorizing, or otherwise nonlocal compatibility structure. It claims neither total tensor-factor separability nor Bell-local factorization. A local realization act may occur inside a standing lineage while tensor-level compatibility constraints remain nonlocal in the Bell-classification sense, as developed in the boundary-level compatibility analysis [9]. Bounded standing localization is therefore not a mechanism for Bell correlations and not a restoration of separable local realism.

15 What remains downstream

This paper supplies a pre-metric locality floor. Several stronger structures remain to be derived in later papers.

Downstream target	Additional work still needed
Neighborhood structure	Show how bounded localization classes induce adjacency or compatibility relations among loci.
Metricity	Show how stability, comparison, and reconstruction constraints induce metric-like structure rather than assuming it.
Finite propagation	Show that perturbation transport between loci is bounded by admissible continuation constraints.
Field carriers	Show that field-like structures are minimal carriers of localized standing persistence.
Particle identity	Show that particle-like identity is a stable localized persistence class rather than primitive objecthood.
Gauge structure	Show that boundary-trace compatibility and role-anchor fixation induce gauge-like redundancy and connection behavior.
Vacuum structure	Show that the minimal concrete persistence regime is non-null without being matter or occupancy.

The present theorem intentionally stops before these stronger conclusions. This restraint is

not a weakness of the result; it is the condition that keeps the derivation non-circular. Metricity, propagation, fields, and particles should not be smuggled into the first locality theorem.

16 Conclusion

The present analysis identifies bounded standing-localization as the structural origin of locality in the realized-physics arc. Paper 1 established that realized persistence cannot be reconstruction-null. Paper 2 adds that non-vacuous standing-bearing persistence cannot retain consequence under perturbation and lawful restoration unless it factors through a bounded continuation-sensitive locus.

Global standing, diffuse standing, trace-only succession, hidden reservoirs, default representatives, external selectors, identity resets, and domain drift do not preserve the relevant lineage inside the same fixed regime. Restorative continuation constraints therefore require a bounded identity locus carrying perturbation burden, restoration, identity-memory, retained standing, quotient stability, boundary role, and fail-closed failure. The induced equivalence class is bounded standing-localization.

The physical role of the result is that locality is already present at this level as a role, though not yet as metric geometry or spacetime structure. Later theories may represent this role by regions, neighborhoods, events, local algebras, field supports, or particle positions. Those are downstream specializations. The present theorem derives the locality role they must preserve.

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